

DAVID HYUNYOO JANG

✉ hwjang00@snu.ac.kr  daylight00  daylight-00 | US Citizen

EDUCATION

Seoul National University

PharmD

Selected Coursework: Biochemistry · Pharmaceutical Statistics · Drug Delivery Systems

Mar 2022 – Feb 2028 (Expected)

Seoul, South Korea

Korea University

Coursework in Materials Science and Engineering (no degree)

Selected Coursework: Organic Nanomaterials · Fundamentals of Data Science

Mar 2019 – Aug 2021

Seoul, South Korea

RESEARCH EXPERIENCE

AI Protein Design Lab, Korea Advanced Institute of Science and Technology (KAIST)

Undergraduate Researcher (advisor: Gyu Rie Lee)

Jun 2025 – Aug 2025

Daejeon, South Korea

Collaborative small-molecule sensor binder design campaigns

- Developed, standardized, and released a reproducible end-to-end binder design workflow during phenytoin and carbamazepine sensor-design campaigns, integrating diffusion-based backbone generation, LigandMPNN, Rosetta scoring, AF3/Boltz-2 refolding, PLACER, TMalign, and RMSD-based triage with parquet-based metric lineage and SLURM-scalable execution. [\[code\]](#)
- Developed metric-guided LigandMPNN–FastRelax recycling to diversify and optimize candidate pools beyond sequence-only sampling across phenytoin/carbamazepine campaigns; in the carbamazepine campaign, expanded screened candidates (21→275) and improved the best Rosetta interface $\Delta\Delta G$ score (−25.4→−31.6). [\[code\]](#)

RFdiffusion2 atom-wise RASA conditioning for ligand orientation control

- Implemented inference-time atom-wise RASA conditioning for RFdiffusion2 to control buried/exposed ligand atoms and improve ligand-orientation control while preserving compatibility with global RASA inputs. [\[PR\]](#)

Laboratory of Computational Drug Discovery, Seoul National University

Undergraduate Researcher (advisor: Juyong Lee)

Jul 2023 – Aug 2025

Seoul, South Korea

PREpiBind — pMHC-II binding: representation benchmark & web server

- Independently built PREpiBind, an open framework benchmarking 10 protein representations — BLOSUM62, structure-prediction embeddings extracted by modifying AlphaFold3/Chai-1/Boltz-1 inference code, and protein language models (ESM C/ESM3) — for pMHC class II binding on 205K IEDB entries; PLM embeddings consistently outperformed structure and alignment representations (counter to the structure-first hypothesis) and matched or exceeded NetMHCIIpan-4.3, MixMHC2pred-2, and DeepNeo.
- Established leakage-controlled splits and stress-tested generalization under leave-one-molecule-out and cross-species (mouse) validation, where PLM embeddings generalized best among the benchmarked representations; re-implemented DeepNeo from scratch in PyTorch (matching its reported accuracy) for a fair head-to-head.
- Selected ESM C 300M for the best accuracy–efficiency trade-off and released open tools — pre-trained models, Colab/CLI demos, and a public web server covering 7,282 IPD-IMGT/HLA alleles zero-shot plus a custom-sequence mode for non-human alleles — with inference optimized to up to ~69× faster than command-line tools. [\[server; code\]](#)

LLMs on the Korean Pharmacist Licensing Exam

- Designed and built an end-to-end benchmark of 27 LLMs — 7 proprietary and 20 open-source (4–104B) — across six KPLE exam years (2019–2024) in Korean and English, covering question extraction (text + vision-OCR), LLM-based translation, local/API inference, and official scoring with human-percentile normalization.
- 7 models passed all years in both languages — the top reaching the top 12% of human examinees — under a logarithmic size–performance trend; developed a multi-pronged data-contamination analysis arguing against pretraining leakage as the dominant performance driver. [\[code\]](#)

CB1 receptor small-molecule virtual screening

- Reproduced an end-to-end structure-based virtual screen of the Enamine Hit Locator Library (460K compounds) against the CB1 receptor on an 8-GPU cluster (~12 h) — self-compiling AutoDock-GPU from source for multi-GPU execution (the prebuilt binary used 1 of 8 GPUs) — and automating batch docking, result parsing, and ranking.

Structure & Knowledge Injection into Machine Learning Lab, Seoul National University

Undergraduate Researcher (advisor: Jay-Yoon Lee)

May 2024 – Sep 2024

Seoul, South Korea

Semi-supervised data augmentation for RoseTTAFold2NA collaboration with Minkyung Baek's lab

- Implemented the training pipeline for a semi-supervised joint-loss objective that augments RF2NA with unlabeled self-distilled structures — built the mixed labeled/unlabeled data loader, per-batch loss routing, and a distributed sampler with tunable unlabeled fraction, consolidating scattered prototype scripts into one parameterized training entry point.

- Designed and ran a controlled low-label study — PDB-only vs distilled-augmentation vs joint-loss semi-supervision across 10/30/50% label-density regimes — to test whether unlabeled data improves RF2NA learning under sparse supervision.

Systems Pharmacology Lab, Seoul National University

Jun 2023 – Jul 2023

Undergraduate Internship (advisor: Minsoo Noh)

Seoul, South Korea

- Performed HaCaT cell culture and PGE₂ competitive ELISA to assess anti-inflammatory effects of tryptophol; conducted dose-response analysis and benchmarking against NSAIDs.

MANUSCRIPTS & PRESENTATIONS

- PREpiBind: Protein Representation-integrated Epitope-MHC Class II Binding Prediction**
David Hyunyoo Jang, Dongwoo Kim, Byungho Park, Untaek Hwang, Yoonjoo Choi, Juyong Lee
KIDDS 2025 Summer Workshop (Korean In silico bioDesign and Discovery Society, Poster)
Manuscript in preparation
- Proprietary and Open-Source Large Language Models on the Korean Pharmacist Licensing Examination: A Comparative Benchmarking Study**
David Hyunyoo Jang, Juyong Lee
Under review
- DeepXeno: Structure-Informed Prediction of Xenotransplantation Immunogenicity**
David Hyunyoo Jang, Byungho Park, Untaek Hwang, Juyong Lee
Outstanding Research Presentation in 2024 SNU Student-Directed Education Undergraduate Research Program (Talk)

AWARDS & FUNDING

Research Excellence Award, 2024 SNU Student-Directed Education Undergraduate Research Program 2025

President of Seoul National University

- Awarded to two projects per fields (ten total) for excellence in research execution and outcomes.

2024 SNU Student-Directed Education Undergraduate Research Program 2024

Faculty of Liberal Education, Seoul National University

- Selected as one of 36 projects to receive funding for undergraduate research. (KRW 7M)

LEADERSHIP & COMMUNICATION

SNU Alpha | Deep Learning Study Club

Spring 2022 – Winter 2024

Executive Member, Co-Founder

College of Pharmacy, Seoul National University

- Peer-taught deep learning fundamentals and applications in computational drug discovery to pharmacy students.
- Led weekly study sessions on Natural Language Processing (Fall 2023) and PyTorch (Spring 2024).

Military Service

Aug 2025 – Feb 2027 (Expected)

Squad Leader, Sergeant

III Corps, Republic of Korea Army

- Led a 10-person squad at an air-defense emplacement, coordinating daily operations and training.
- Operated and maintained air-defense radar systems in a frontline sector along the Korean DMZ.

Seoul National University Hospital

Dec 2024 – Feb 2025

Hospital Pharmacy Practicum

Seoul, South Korea

- Completed rotations in 12 specialty departments, gaining broad exposure to hospital pharmacy practice.
- Conducted drug therapy case reviews and provided patient counseling in multidisciplinary clinical settings.

E-Blue Pharmacy

Feb 2025 – Mar 2025

Community Pharmacy Practicum

Suwon, South Korea

- Screened prescriptions for errors and performed medication therapy management interventions.
- Conducted patient counseling for prescription medications and over-the-counter consultations.

TECHNICAL SKILLS

Programming & Data: Python, PyTorch, Numpy, Bash, pandas, matplotlib, Streamlit, Nginx

Modeling & Docking: RDKit, PyMOL, Chimera, AutoDock, PyRosetta, RFdiffusion (AA), LigandMPNN, AlphaFold3

Systems & DevOps: Linux, Git, Slurm, conda/mamba, HuggingFace

Experimental Techniques: cell culture (HaCaT), buffer/reagent prep, competitive ELISA, serial dilutions, plate reader